United States Patent Application Publication 20250275213
Kind Code A1
Publication Date August 28, 2025
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MEMORY DEVICE WITH CONTAINER-SHAPED ELECTRODE AND METHOD FOR FABRICATING THE SAME

Abstract

The present application discloses a memory device and a method for fabricating the memory device. The memory device includes a substrate; a landing area positioned on the substrate; a bottom electrode positioned on the landing area, wherein the bottom electrode has a container-shaped profile; a support layer positioned over the substrate and laterally surrounded the bottom electrode; a dielectric structure including a dielectric layer conformally positioned on the bottom electrode and on a top surface of the support layer, and covering top corners of the support layer, and a plurality of dielectric portions conformally positioned on the dielectric layer and covering the top corners of the support layer; and a top electrode structure positioned on the dielectric structure. The dielectric portions are sandwiched by the top electrode structure and the dielectric layer. The top surface of the third support layer is higher than a top surface of the bottom electrode.

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Family ID: **1000007828243**

Appl. No.: **18/635355**

Filed: **April 15, 2024**

Related U.S. Application Data

parent US division 18584150 20240222 PENDING child US 18635355

Publication Classification

Int. Cl.: **H01L29/41** (20060101); **H01L21/28** (20250101); **H01L29/06** (20060101); **H10B12/00** (20230101)

U.S. Cl.:

CPC **H10D64/20** (20250101); **H10B12/31** (20230201); **H10D62/115** (20250101); H10D64/031 (20250101)

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application is a continuation application of U.S. Non-Provisional application Ser. No. 18/584,150 filed Feb. 22, 2024, which is incorporated herein by reference in its entirety.

TECHNICAL FIELD
[0002] The present disclosure relates to a memory device and a method for fabricating the memory device, and more particularly, to a memory device with a container-shaped electrode and a method for fabricating the memory device with the container-shaped electrode.
DISCUSSION OF THE BACKGROUND
[0003] Continuous advancements in technology have allowed capacitors to have increasingly high aspect ratios in fabrication of integrated circuits. A storage electrode of a memory device is shaped as a container. The advancements in technology are resulting in increasing heights of container-shaped storage nodes.
[0004] However, conventional etching techniques (particularly techniques for high-aspect-ratio etching) limit performance of memory devices. Specifically, a recess formed by the conventional etching techniques has a smaller dimension at a bottom, reducing a capacitance of a capacitor formed in the recess. In addition, some types of conventional wet etch techniques may cause significant material loss and device defects due to poor control.
[0005] This Discussion of the Background section is provided for background information only. The statements in this Discussion of the Background are not an admission that the subject matter disclosed in this Discussion of the Background section constitute prior art to the present disclosure, and no part of this Discussion of the Background section may be used as an admission that any part of this application, including this Discussion of the Background section, constitutes prior art to the present disclosure.
SUMMARY
[0006] One aspect of the present disclosure provides a memory device including a substrate; a landing area positioned on the substrate; a bottom electrode positioned on the landing area, wherein the bottom electrode has a container-shaped profile; a support layer positioned over the substrate and laterally surrounded the bottom electrode; a dielectric structure including a dielectric layer conformally positioned on the bottom electrode and on a top surface of the support layer, and covering top corners of the support layer, and a plurality of dielectric portions conformally positioned on the dielectric layer and covering the top corners of the support layer; and a top electrode structure positioned on the dielectric structure. The plurality of dielectric portions are sandwiched by the top electrode structure and the dielectric layer. The top surface of the third support layer is higher than a top surface of the bottom electrode.
[0007] Another aspect of the present disclosure provides a memory device including a substrate; a landing area positioned on the substrate; a first support layer positioned on the substrate; a second support layer positioned over the first support layer; a third support layer positioned over the second support layer; a bottom electrode including a bottom portion horizontally positioned on the landing area, a first wall portion extending from the bottom portion along a first direction, a second wall portion extending from the bottom portion along the first direction and separated from the first wall portion, and a top portion connecting to the second wall portion, parallel to the bottom portion,

and collectively covering a top corner of the third support layer with the second wall portion; a dielectric structure including a dielectric layer conformally positioned on the bottom electrode, and a dielectric portion positioned on the dielectric layer and covering the top corner of the third support layer; and a top electrode conformally positioned on the dielectric structure.

[0008] Another aspect of the present disclosure provides a method for fabricating a memory device including forming a landing area on a substrate; forming an energy-removable layer adjacent to the landing area; sequentially forming a first support layer, a first material layer, a second material layer, a second support layer, a third material layer, and a third support layer over the landing area, wherein the first material layer is doped with an N-type dopant and the third material layer is doped with a P-type dopant; forming a first recess to expose the landing area; forming a bottom electrode within the first recess; conformally forming a dielectric layer on the bottom electrode and on a top surface of the third support layer; forming a plurality of dielectric portions on top corners of the first recess; and conformally forming a top conductive layer on the dielectric layer and covering the plurality of dielectric portions; and forming a top electrode on the top conductive layer.

[0009] Due to the design of the memory device of the present disclosure, the overall thickness of the dielectric structure at the top corner of the third support layer is increased. As a result, the leakage current between the bottom electrode and the top electrode structure can be prevented or reduced. In addition, the first width of the lower portion of the top electrode structure is greater than the third width of the upper portion of the top electrode structure. Consequently, the capacitance of the capacitor structure can be improved. Furthermore, the air gap structure positioned between the plurality of landing areas can reduce the parasitic capacitance between the plurality of landing areas.

[0010] The foregoing has outlined rather broadly the features and technical advantages of the present disclosure in order that the detailed description of the disclosure that follows may be better understood. Additional features and advantages of the disclosure will be described hereinafter, and form the subject of the claims of the disclosure. It should be appreciated by those skilled in the art that the conception and specific embodiment disclosed may be readily utilized as a basis for modifying or designing other structures or processes for carrying out the same purposes of the present disclosure. It should also be realized by those skilled in the art that such equivalent constructions do not depart from the spirit and scope of the disclosure as set forth in the appended claims.

Description

BRIEF DESCRIPTION OF THE DRA WINGS

[0011] Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It should be noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0012] FIG. **1** is a cross-sectional view of a semiconductor device in accordance with a comparative embodiment;

[0013] FIG. **2** is a flow diagram illustrating a method of manufacturing a memory device in accordance with some embodiments of the present disclosure;

[0014] FIGS. **3** to **13** are cross-sectional diagrams of intermediate stages in the formation of the memory device in accordance with some embodiments of the present disclosure;

[0015] FIG. **14** is a top view perspective schematic diagram of an intermediate stage in the formation of the memory device shown in FIG. **13** in accordance with some embodiments of the present disclosure;

[0016] FIGS. **15** to **25** are cross-sectional diagrams of intermediate stages in the formation of the

memory device in accordance with some embodiments of the present disclosure;
[0017] FIG. **26** is a flow diagram illustrating a method of manufacturing a memory device in accordance with some embodiments of the present disclosure;
[0018] FIG. **27** is a cross-sectional diagram of an intermediate stage in the formation of the memory device in accordance with some embodiments of the present disclosure;
[0019] FIG. **28** is a top view perspective schematic diagram of an intermediate stage in the formation of the memory device shown in FIG. **27** in accordance with some embodiments of the present disclosure; and
[0020] FIGS. **29** to **34** are cross-sectional diagrams of intermediate stages in the formation of the memory device in accordance with some embodiments of the present disclosure.

DETAILED DESCRIPTION

[0021] The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.
[0022] Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.
[0023] It should be understood that when an element or layer is referred to as being “connected to” or “coupled to” another element or layer, it can be directly connected to or coupled to another element or layer, or intervening elements or layers may be present.
[0024] It should be understood that, although the terms first, second, etc. may be used herein to describe various elements, these elements should not be limited by these terms. Unless indicated otherwise, these terms are only used to distinguish one element from another element. Thus, for example, a first element, a first component or a first section discussed below could be termed a second element, a second component or a second section without departing from the teachings of the present disclosure.
[0025] Unless the context indicates otherwise, terms such as “same,” “equal,” “planar,” or “coplanar,” as used herein when referring to orientation, layout, location, shapes, sizes, amounts, or other measures do not necessarily mean an exactly identical orientation, layout, location, shape, size, amount, or other measure, but are intended to encompass nearly identical orientation, layout, location, shapes, sizes, amounts, or other measures within acceptable variations that may occur, for example, due to manufacturing processes. The term “substantially” may be used herein to reflect this meaning. For example, items described as “substantially the same,” “substantially equal,” or “substantially planar,” may be exactly the same, equal, or planar, or may be the same, equal, or planar within acceptable variations that may occur, for example, due to manufacturing processes.
[0026] It should be noted that, in the description of the present disclosure, above (or up) corresponds to the direction of the arrow of the direction Z, and below (or down) corresponds to the opposite direction of the arrow of the direction Z.

[0027] FIG. **1** is a cross-sectional view of a semiconductor device in accordance with a comparative embodiment. For example, as shown in FIG. **1**, a recess R**0** is formed in an insulation stack, wherein the insulation stack includes a borophosphosilicate glass (BPSG) layer **997** and an oxide layer **998** over the BPSG layer **997**. An electrode **999** is formed in the recess R**0**. However, a dimension K**1** at a top of the recess R**0** is substantially greater than a dimension K**2** at a bottom of the recess R**0**. For example, the dimension K**1** may be greater than the dimension K**2** by 10 nm to 15 nm. Smaller values of the dimension K**2** lead to lower capacitance of a capacitor further formed in the recess R**0**. Alternatively stated, a capacitance of a capacitor structure formed in the recess R**0** may be severely limited due to a limitation of the dimension K**2** at the bottom of the recess R**0**.
[0028] In addition, conventional wet etch techniques may cause significant material loss due to poor etch rate control. Alternatively stated, it is difficult to control a profile of the recess R**0**; for example, a bowing issue may occur at a position proximal to the top of the recess R**0**. Such issues may lead to poor device performance, or even cause electrical shorting between electrodes.
[0029] In order to meet cell capacitance requirements for advanced memory devices, such as dynamic random-access memory (DRAM) devices, a method for increasing capacitance is required. Accordingly, there is a need to develop a new memory device and a new method for forming the same. A first embodiment is depicted in FIGS. **2** to **25**, and a second embodiment is depicted in FIGS. **26** to **34**, wherein the second embodiment may further include operations depicted in FIGS. **3** to **15** that can be performed prior to the operations depicted in FIGS. **27** to **34**.
[0030] FIG. **2** is a flow diagram illustrating a method S**1** of manufacturing a memory device in accordance with some embodiments of the present disclosure. The method S**1** includes a number of operations (S**11**, S**12**, S**13**, S**14**, S**15**, S**16**, S**17**, S**18**, S**19**, S**20**, S**21**, and S**22**) and the description and illustration are not deemed as a limitation to the sequence of the operations. In the operation S**11**, a landing area is formed over a substrate, an energy-removable layer is formed adjacent to the landing area, and a first support layer is formed over the landing area. In the operation S**12**, a first material layer is formed over the first support layer, wherein the first material layer is doped with an N-type dopant. In the operation S**13**, a second material layer is formed over the first material layer. In the operation S**14**, a second support layer is formed over the second material layer. In the operation S**15**, a third material layer is formed over the second support layer, wherein the third material layer is doped with a P-type dopant. In the operation S**16**, a third support layer is formed over the third material layer. In the operation S**17**, a dry etch operation is performed to form a first recess, wherein the first recess penetrates the first material layer, the second material layer, and the third material layer. In the operation S**18**, a bottom electrode material layer is formed in the first recess. In the operation S**19**, a portion of the bottom electrode material layer is removed to form a bottom electrode. In the operation S**20**, a dielectric layer is conformally formed over the bottom electrode and a plurality of dielectric portions are formed on the dielectric layer and covering top corners of the first recess. In the operation S**21**, a top conductive layer and a top electrode are formed over the dielectric layer and the plurality of dielectric portions. In the operation S**22**, an air gap structure is formed by performing a thermal treating process.
[0031] FIGS. **3** to **25** are schematic diagrams illustrating various fabrication stages constructed according to the method S**1** in accordance with some embodiments of the present disclosure.
[0032] FIG. **3** is a cross-sectional diagram of an intermediate stage in the formation of a memory device in accordance with some embodiments of the present disclosure. In the operation S**11**, a substrate **101** is provided. An insulation layer **103** is formed over the substrate **101**, and a plurality of landing areas **102** are formed in the insulation layer **103**. For example, a landing area **102** is formed over a first region RA of the substrate **101**, and another landing area **102** is formed over a second region RB that is different from the first region RA. In some embodiments, a chemical mechanical planarization (CMP) operation can be performed from above the insulation layer **103** and the landing areas **102**. In some embodiments, the landing areas **102** include conductive material.

[0033] The substrate **101** may be a semiconductor substrate, such as a bulk semiconductor, a semiconductor-on-insulator (SOI) substrate, or the like. The substrate **101** can include an elementary semiconductor including silicon or germanium in a single crystal form, a polycrystalline form, or an amorphous form; a compound semiconductor material including at least one of silicon carbide, gallium arsenide, gallium phosphide, indium phosphide, indium arsenide, and indium antimonide; an alloy semiconductor material including at least one of silicon, SiGe, GaAsP, AlInAs, AlGaAs, GaInAs, GaInP, and GaInAsP; any other suitable materials; or a combination thereof. In some embodiments, the alloy semiconductor substrate may be a silicon alloy with a gradient silicon feature in which Si and metal compositions change from one ratio at one location to another ratio at another location of the gradient silicon feature. For example, the alloy semiconductor substrate may be a SiGe alloy with a gradient SiGe feature in which Si and Ge compositions change from one ratio at one location to another ratio at another location of the gradient SiGe feature. In another embodiment, the SiGe alloy is formed over a silicon substrate. In some embodiments, a SiGe alloy can be mechanically strained by another material in contact with the SiGe alloy.
[0034] In some embodiments, the substrate **101** may have a multilayer structure, or the substrate **101** may include a multilayer compound semiconductor structure. In some embodiments, the substrate **101** includes semiconductor devices, electrical components, electrical elements, or a combination thereof. In some embodiments, the substrate **101** includes transistors or functional units of transistors.
[0035] FIG. **4** is a cross-sectional diagram of an intermediate stage in the formation of a memory device in accordance with some embodiments of the present disclosure. A first mask layer **996** is formed on the insulation layer **103**. The first mask layer **996** is a photoresist layer and has a pattern exposing a portion of the insulation layer **103** between the landing area **102** formed over the first region RA and the landing area **102** formed over the second region RB.
[0036] FIG. **5** is a cross-sectional diagram of an intermediate stage in the formation of a memory device in accordance with some embodiments of the present disclosure. A first opening OP**1** is formed penetrating the insulation layer **103** by performing a dry etch operation. In some embodiments, the dry etch operation may include applying plasma, such as fluorine-based plasma or fluorine-containing plasma. In some embodiments, a cleaning operation can be performed to remove residues generated in the dry etch operation. After the formation of the first opening OP**1**, the first mask layer **996** is removed.
[0037] FIG. **6** is a cross-sectional diagram of an intermediate stage in the formation of a memory device in accordance with some embodiments of the present disclosure. In the operation S**11**, an energy-removable layer **213** is formed in the first opening OP**1**. A planarization operation is performed to provide a substantially flat surface for subsequent operations.
[0038] In some embodiments, the energy-removable layer **213** includes a thermal decomposable material. In some other embodiments, the energy-removable layer **213** includes a photonic decomposable material, an e-beam decomposable material, or another applicable energy decomposable material. Detailedly, in some embodiments, the energy-removable layer **213** includes a base material and a decomposable porogen material that is substantially removed once being exposed to an energy source (e.g., heat).
[0039] In some embodiments, the base material includes hydrogen silsesquioxane (HSQ), methylsilsesquioxane (MSQ), porous polyarylether (PAE), porous SiLK, or porous silicon oxide (SiO.sub.2), and the decomposable porogen material includes a porogen organic compound, which can provide porosity to the space originally occupied by the energy-removable layer **213** in the subsequent operations.
[0040] FIG. **7** is a cross-sectional diagram of an intermediate stage in the formation of a memory device in accordance with some embodiments of the present disclosure. In the operation S**11**, a first support layer **104** is formed over the substrate **101**. The first support layer **104** extends over the

insulation layer **103**, the energy-removable layer **213**, and the landing areas **102**. In some embodiments, a material of the first support layer **104** includes silicon nitride (SiN). Alternatively, a material of the first support layer **104** can be another suitable material, such as an insulation material.
[0041] FIG. **8** is a cross-sectional diagram of an intermediate stage in the formation of a memory device in accordance with some embodiments of the present disclosure. In the operation S**12**, a first material layer **105** is formed over the first support layer **104**, wherein the first material layer **105** is doped with an N-type dopant. For example, the first material layer **105** can be doped with phosphorus, arsenic, antimony, bismuth, or other suitable dopants. In some embodiments, the first material layer **105** includes polysilicon doped with an N-type dopant. The first material layer **105** has a first thickness h**1**. In some embodiments, the first thickness h**1** of the first material layer **105** is less than 400 nm. In some embodiments, when the first material layer **105** is doped with phosphorus, a concentration of phosphorus dopant in the first material layer **105** is in a range from 5E14 atoms/cm.sup.3 to about 5E15 atoms/cm3.
[0042] FIG. **9** is a cross-sectional diagram of an intermediate stage in the formation of a memory device in accordance with some embodiments of the present disclosure. In the operation S**13**, a second material layer **106** is formed over the first material layer **105**. The second material layer **106** has a second thickness h**2**. In some embodiments, the second thickness h**2** of the second material layer **106** is less than the first thickness h**1** of the first material layer **105**. In some embodiments, the second thickness h**2** of the second material layer **106** is less than 200 nm. In some embodiments, the second material layer **106** includes undoped polysilicon (which can be referred to as intrinsic polysilicon), that is, without significant dopant therein. The second material layer **106** may be in direct contact with the first material layer **105**.
[0043] FIG. **10** is a cross-sectional diagram of an intermediate stage in the formation of a memory device in accordance with some embodiments of the present disclosure. In the operation S**14**, a second support layer **107** is formed over the second material layer **106**. A material of the second support layer **107** may be identical to the material of the first support layer **104**. In some embodiments, a material of the second support layer **107** includes silicon nitride (SiN).
[0044] FIG. **11** is a cross-sectional diagram of an intermediate stage in the formation of a memory device in accordance with some embodiments of the present disclosure. In the operation S**15**, a third material layer **108** is formed over the second support layer **107**, wherein the third material layer **108** is doped with a P-type dopant. For example, the third material layer **108** can be doped with boron, indium, gallium, or other suitable dopants. The third material layer **108** has a third thickness h**3**. In some embodiments, the third thickness h**3** of the third material layer **108** is greater than the first thickness h**1** of the first material layer **105**. In some embodiments, the third thickness h**3** of the third material layer **108** is greater than the second thickness h**2** of the second material layer **106**. In some embodiments, the third thickness h**3** of the third material layer **108** is less than 500 nm. In some embodiments, a sum of the first thickness h**1** of the first material layer **105** and the second thickness h**2** of the second material layer **106** is greater than the third thickness h**3** of the third material layer **108**.
[0045] In some embodiments, the third material layer **108** includes polysilicon doped with a P-type dopant. In some embodiments, when the third material layer **108** is doped with boron, a concentration of boron dopant in the third material layer **108** is in a range from 5E14 atoms/cm.sup.3 to about 5E15 atoms/cm.sup.3. The third material layer **108** may be in direct contact with the second support layer **107**. The third material layer **108** has a top surface **108**T**1**.
[0046] FIG. **12** is a cross-sectional diagram of an intermediate stage in the formation of a memory device in accordance with some embodiments of the present disclosure. In the operation S**16**, a third support layer **109** is formed over the top surface **108**T**1** of the third material layer **108**. A material of the third support layer **109** may be identical to the material of the first support layer **104** or the material of the second support layer **107**. In some embodiments, a material of the third

support layer **109** includes silicon nitride (SiN). In some embodiments, a total thickness h**4** of a stack of the first support layer **104**, the first material layer **105**, the second material layer **106**, the second support layer **107**, the third material layer **108**, and the third support layer **109** may be in a range from 0.8 μm to about 1.2 μm.

[0047] FIG. **13** is a cross-sectional diagram of an intermediate stage in the formation of a memory device in accordance with some embodiments of the present disclosure, and FIG. **14** is a top view perspective schematic diagram of an intermediate stage in the formation of a memory device shown in FIG. **13** in accordance with some embodiments of the present disclosure. In the operation S**17**, a plurality of first recesses R**1** are formed over the substrate **101** by performing a dry etch operation. In some embodiments, the dry etch operation may include applying plasma, such as fluorine-based plasma or fluorine-containing plasma. In some embodiments, each of the plurality of first recesses R**1** penetrates the first support layer **104**, the first material layer **105**, the second material layer **106**, the second support layer **107**, the third material layer **108**, and the third support layer **109**. The plurality of landing areas **102** are exposed through the plurality of first recesses R**1**. In some embodiments, one landing area **102** corresponds to one first recess R**1**. In some embodiments, positions of the plurality of first recesses R**1** are defined by one or more lithography operation(s), and a cleaning operation can be performed to remove residues generated in the dry etch operation. In some embodiments, as depicted in FIG. **14**, the plurality of first recesses R**1** may be arranged in a staggered array. A position of each of the plurality of first recesses R**1** corresponds to a position of one of the plurality of landing areas **102**.

[0048] Each first recess R**1** has a first portion P**1** laterally surrounded by the first material layer **105**, a second portion P**2** laterally surrounded by the second material layer **106**, a third portion P**3** laterally surrounded by the third material layer **108**, and a fourth portion P**4** laterally surrounded by the first support layer **104**. The third portion P**3** is above the second portion P**2**, the second portion P**2** is above the first portion P**1**, and the first portion P**1** is above the fourth portion P**4**.

[0049] An etch rate of the dry etch operation in the operation S**17** on the first material layer **105** is greater than an etch rate of the dry etch operation on the second material layer **106**. Further, the etch rate of the dry etch operation on the second material layer **106** is greater than an etch rate of the dry etch operation on the third material layer **108**. Detailedly, by doping the first material layer **105** with an N-type dopant in a predetermined concentration, the etch rate of the dry etch operation in the operation S**17** on the first material layer **105** in a lateral direction can be enhanced. In contrast, by doping the third material layer **108** with a P-type dopant in a predetermined concentration, the etch rate of the dry etch operation in the operation S**17** on the third material layer **108** in a lateral direction can be reduced.

[0050] In some embodiments, an aspect ratio of one first recess R**1** may be greater than 35. The present disclosure seeks to address an issue of the reduced dimension at the bottom of the recess in the comparative embodiment depicted in FIG. **1**. That is, a recess formed by conventional techniques may result in a greater difference between a top dimension and a bottom dimension, which limits device performance. In contrast, a configuration of the first material layer **105**, the second material layer **106**, and the third material layer **108** can mitigate such issue and improve device performance.

[0051] Alternatively stated, by doping the first material layer **105** and the third material layer **108** with different types of dopants and disposing the second material layer **106** without significant doping between the first material layer **105** and the third material layer **108**, profiles of the first recesses R**1** can be improved.

[0052] Referring to FIG. **13**, in some embodiments, a dimension CD**1** of the first portion P**1** is greater than a dimension CD**2** of the second portion P**2**. In some embodiments, the dimension CD**2** of the second portion P**2** is greater than a dimension CD**3** of the third portion P**3**. In some embodiments, the dimension CD**1** of the first portion P**1** is greater than a dimension CD**4** of the fourth portion P**4**. In some embodiments, the dimension CD**1** is greater than either the dimension

CD**2** or the dimension CD**3**. The aforementioned ranges of dopant concentrations in the material layers are designed to obtain desired values for the dimensions CD**1**, CD**2** and CD**3**, with a difference between CD**1** and CD**3** within a predetermined range.

[0053] In some alternative embodiments, the dimension CD**1** is comparable to the dimension CD**2**, and the dimension CD**2** is comparable to the dimension CD**3**.

[0054] In some embodiments, a top surface **102**T of each of the landing areas **102** is exposed through the fourth portion P**4** of the first recess R**1**. A portion of each of the landing areas **102** may be covered by the first support layer **104**.

[0055] FIG. **15** is a cross-sectional diagram of an intermediate stage in the formation of a memory device in accordance with some embodiments of the present disclosure. In the operation S**18**, a bottom electrode material layer **111**M is formed in the plurality of first recesses R**1**, thus causing the bottom electrode material layer **111**M to have a container-shaped profile. In the present disclosure, the term “container-shaped” or “shaped as a container” may be referred to an object having a bottom and a sidewall portion extended from the bottom, and a portion of a space above the bottom is at least partially surrounded by the sidewall portion in lateral direction. The bottom electrode material layer **111**M conforms to a profile of the first recess R**1**. In some embodiments, the bottom electrode material layer **111**M further covers a top surface **109**T of the third support layer **109** and the top surface **102**T of each of the landing areas **102**. In some embodiments, the bottom electrode material layer **111**M may include titanium nitride (TiN) and may be formed by blanket deposition, chemical vapor deposition (CVD), physical vapor deposition (PVD), atomic layer deposition (ALD), or the like.

[0056] After the forming of the bottom electrode material layer **111**M, the first portion P**1** of the first recess R**1** has a dimension D**1**, the second portion P**2** of the first recess R**1** has a dimension D**2**, and the third portion P**3** of the first recess R**1** has a dimension D**3**. In some embodiments, the dimension D**3** of the third portion P**3** is less than the dimension D**2** of the second portion P**2**. In some embodiments, the dimension D**2** of the second portion P**2** is less than the dimension D**1** of the first portion P**1**. In some embodiments, the dimension D**1** is greater than either the dimension D**2** or the dimension D**3**. In some embodiments, the dimension D**3** is in a range from about 26 nm to about 36 nm in certain technology nodes, but the present disclosure is not limited thereto. In some embodiments, a difference between the dimension D**3** and the dimension D**1** is less than 10 nm, or in some cases, less than 5 nm.

[0057] In some alternative embodiments, the dimension D**1** is comparable to the dimension D**2**, and the dimension D**2** is comparable to the dimension D**3**.

[0058] In some embodiments, the dimensions D**1**, D**2** and D**3** represent a maximum width in a horizontal direction of an empty space of the first portion P**1**, the second portion P**2**, and the third portion P**3**, respectively, of the first recess R**1**.

[0059] FIG. **16** is a cross-sectional diagram of an intermediate stage in the formation of a memory device in accordance with some embodiments of the present disclosure. A sacrificial material **137** is formed over the substrate **101** and fills the plurality of first recesses R**1**. In some embodiments, the bottom electrode material layer **111**M is completely covered by the sacrificial material **137**. In some embodiments, the sacrificial material **137** includes an oxide, such as silicon oxide. However, other materials, such as silicide nitride, silicon oxynitride, silicon carbonitride, combinations thereof, or the like, may alternatively be used for the sacrificial material **137**. In some embodiments, the sacrificial material **137** is formed by a deposition process, such as a CVD process, a PVD process, or an ALD process.

[0060] FIG. **17** is a cross-sectional diagram of an intermediate stage in the formation of a memory device in accordance with some embodiments of the present disclosure. An etch process is subsequently performed on the sacrificial material **137** to expose the bottom electrode material layer **111**M, leaving a portion of the sacrificial material **137** in the plurality of first recesses R**1**, which is referred to as a sacrificial structure **137**′. In the present embodiment, a portion of the sacrificial material **137** in the plurality of first recesses R**1** is also removed such that a plurality of recesses **142** are formed over the sacrificial structure **137**′.

[0061] In some embodiments, the top surface **111**MT of the bottom electrode material layer **111**M is higher than the top surface **137**′T of the sacrificial structure **137**′. In some embodiments, the top surface **109**T of the third support layer **109** is higher than the top surface **137**′T of the sacrificial structure **137**′. That is, a portion of the bottom electrode material layer **111**M covering the sidewalls SW of the plurality of first recesses R**1** is exposed in the plurality of the recesses **142** over the sacrificial structure **137**′. The etching process for forming the sacrificial structure **137**′ may be a wet etch process, a dry etch process, or a combination thereof.

[0062] FIG. **18** is a cross-sectional diagram of an intermediate stage in the formation of a memory device in accordance with some embodiments of the present disclosure. In the operation S**19**, an etching process is performed to remove the exposed portions of the bottom electrode material layer **111**M, leaving a portion of the bottom electrode material layer **111**M in the plurality of first recesses R**1**, which is referred to as the plurality of bottom electrodes **111**. In some embodiments, the plurality of bottom electrodes **111** cover the sidewalls SW of the plurality of first recesses R**1** and the top surfaces **102**T of the plurality of landing areas **102**.

[0063] In some embodiments, the portion of the bottom electrode material layer **111**M covering the top corners TC of the plurality of first recesses R**1** is removed, and the top surfaces **111**T**1** of the plurality of bottom electrodes **111** are substantially coplanar with the top surfaces **137**′T of the sacrificial structure **137**′. Within the context of this disclosure, the word “substantially” means preferably at least 90%, more preferably 95%, even more preferably 98%, and most preferably 99%. In other words, each of the plurality of recesses **142** is laterally expanded, in accordance with some embodiments. Moreover, in some embodiments, the top surface **109**T of the third support layer **109**, and the top corners TC, and the upper portion of the sidewalls SW of the plurality of first recesses R**1** are exposed. In some embodiments, the etching process for forming the plurality of bottom electrodes **111** includes one or more dry etching processes.

[0064] FIG. **19** is a cross-sectional diagram of an intermediate stage in the formation of a memory device in accordance with some embodiments of the present disclosure. An etch process is performed to remove the sacrificial structure **137**′. In some embodiments, the etching selectivity of the sacrificial structure **137**′ with respect to the plurality of bottom electrodes **111** is relatively high. Therefore, the sacrificial structure **137**′ is removed by the etching process while the plurality of bottom electrodes **111** may be substantially left. In some embodiments, the etch process for removing the sacrificial structure **137**′ includes a wet etch process.

[0065] FIG. **20** is a cross-sectional diagram of an intermediate stage in the formation of a memory device in accordance with some embodiments of the present disclosure. In the operation S**20**, a dielectric layer **112** is conformally deposited over the top surface **109**T of the third support layer **109** and covering the plurality of bottom electrodes **111**. In some embodiments, the exposed top surfaces **109**T of the third support layer **109** are completely covered by the dielectric layer **112**. Moreover, in some embodiments, the top corners TC of the plurality of first recesses R**1**, the exposed portions of the sidewalls SW of the plurality of first recesses R**1** and the top surfaces **111**T**1** of the plurality of bottom electrodes **111** are covered by the dielectric layer **112**. In some embodiments, the top surfaces **109**T of the third support layer **109**, the top corners TC of the plurality of first recesses R**1**, the exposed portions of the sidewalls SW of the plurality of first recesses R**1**, and the top surfaces **111**T**1** of the plurality of bottom electrodes **111** are in direct contact with the dielectric layer **112**.

[0066] In some embodiments, the dielectric layer **112** includes a single layer or multiple layers. In some embodiments, the dielectric layer **112** includes SiO.sub.2, a dielectric material with high dielectric constant (high-k), such as ZrO.sub.2, HfO.sub.2, TiO.sub.2, AlO, or a combination thereof. For example, the dielectric layer **112** may be a tri-layer structure including two layers of aluminum oxide and a layer of zirconium oxide disposed between them. Moreover, in some embodiments, the dielectric layer **112** is formed by a conformal deposition process, such as a CVD process, a PVD process, or an ALD process.

[0067] FIG. **21** is a cross-sectional diagram of an intermediate stage in the formation of a memory device in accordance with some embodiments of the present disclosure. In the operation S**20**, a plurality of dielectric portions **157** are deposited over the dielectric layer **112** and at the top corners TC of the plurality of first recesses R**1**. In some embodiments, the dielectric layer **112** and the plurality of dielectric portions **157** are collectively referred to the dielectric structure **139**. It should be noted that the plurality of dielectric portions **157** are formed by a deposition process having a lower step coverage than the deposition process used for forming the dielectric layer **112**. In some embodiments, the plurality of dielectric portions **157** are formed by a non-conformal deposition process, such as a non-conformal liner atomic layer deposition (NOLA) process.

[0068] After the plurality of dielectric portions **157** are formed, a portion of the dielectric layer **112** covering the top surface **109**T of the third support layer **109** and a portion of the dielectric layer **112** in the plurality of first recesses R**1** are exposed. In some embodiments, the dielectric layer **112** has a portion sandwiched between the plurality of dielectric portions **157** and the third support layer **109**, and another portion sandwiched between the plurality of dielectric portions **157** and the plurality of bottom electrodes **111**. In some embodiments, the plurality of dielectric portions **157** include silicon oxide, silicon nitride, silicon oxynitride, a dielectric material with high dielectric constant (high-k), or a combination thereof.

[0069] The plurality of dielectric portions **157** are formed to prevent the corner effect, which incur uneven thickness of the dielectric layer **112**. For example, the thickness of the dielectric layer **112** at the top corners TC of the plurality of first recesses R**1** is less than that of other portions of the dielectric layer **112**. By forming the plurality of dielectric portions **157** over the dielectric layer **112**, the overall thickness of the dielectric structure **139** at the top corners TC of the plurality of first recesses R**1** may be increased, which prevent or reduce leakage current between the plurality of bottom electrodes **111** and the subsequently formed top electrode.

[0070] FIG. **22** is a cross-sectional diagram of an intermediate stage in the formation of a memory device in accordance with some embodiments of the present disclosure. In the operation S**21**, a top conductive layer **165** is formed over the dielectric structure **139**. In some embodiments, the top conductive layer **165** is conformally deposited to cover the dielectric layer **112** and the dielectric portions **157** of the dielectric structure **139**.

[0071] Detailedly, the top conductive layer **165** extends along the top surface **109**T of the third support layer **109**, the sidewalls SW of the plurality of first recesses R**1** and the top surfaces **102**T of the plurality of landing areas **102**. In some embodiments, the remaining portion of the plurality of first recesses R**1** is not formed by the top electrode **165**. That is, a portion of the plurality of first recesses R**1** located above the top conductive layer **165**. In some embodiments, the upper portion of the top conductive layer **165** in the plurality of first recesses R**1** is in direct contact with the dielectric portions **157** and the lower portion of the top conductive layer **165** in the plurality of first recesses R**1** is in direct contact with the dielectric layer **112**. Moreover, since the dielectric portions **157** of the dielectric structure **139** do not extend to the bottommost portion of the plurality of first recesses R**1**, the bottommost surface **157**B of each of the dielectric portions **157** is higher than the bottommost surface **165**B of the top conductive layer **165**.

[0072] In some embodiments, the top conductive layer **165** includes titanium nitride (TiN), titanium silicon nitride (TiSiN), or a combination thereof. In some embodiments, the top conductive layer **165** is formed by a deposition process, such as a CVD process or an ALD process.

[0073] FIG. **23** is a cross-sectional diagram of an intermediate stage in the formation of a memory device in accordance with some embodiments of the present disclosure. A top electrode material layer **113**M is formed in the plurality of first recesses R**1**. The top electrode material layer **113**M may include TiN and can be formed by a plating operation or another type of deposition operation. In some embodiments, a portion of the top electrode material layer **113**M is above the third support layer **109**. A second mask layer **995** is formed on the top electrode material layer **113**M. The second mask layer **995** is a photoresist layer and has a pattern covering portions of the top electrode material layer **113**M. The covered portions of the top electrode material layer **113**M are disposed over the plurality of plurality of landing areas **102**, respectively and correspondingly.

[0074] FIG. **24** is a cross-sectional diagram of an intermediate stage in the formation of a memory device in accordance with some embodiments of the present disclosure. In the operation S**21**, a plurality of top electrodes **113** are formed over the plurality of first recesses R**1** by performing a dry etch operation using the second mask layer **995** as the mask. The dry etch operation may stop at the dielectric layer **112**. The top conductive layer **165** is separated into multiple portions and is referred to as the plurality of top conductive layers **165**. The top electrode material layer **113**M is separated into multiple portions and is referred to as the plurality of top electrodes **113**. The plurality of top conductive layers **165** and the plurality of top electrodes **113** are collectively referred to the plurality of top electrode structures **140**. The plurality of bottom electrodes **111**, the dielectric structure **139**, and the plurality of top electrode structures **140** are collectively referred to the plurality of capacitance structures **148**. After the formation of the capacitance structures **148**, the second mask layer **995** is removed.

[0075] In some embodiments, each of the plurality of top electrode structures **140** of the plurality of capacitance structures **148** has a lower portion Q**1** laterally surrounded by the first material layer **105**, a middle portion Q**2** laterally surrounded by the second material layer **106**, and an upper portion Q**3** laterally surrounded by the third material layer **108** and the third support layer **109**. The lower portion Q**1** has a first width W**1** measured from a first inner sidewall SD**1** of the top electrode structure **140** to a second inner sidewall SD**1**′ of the top electrode structure **140**, the middle portion Q**2** has a second width W**2** measured from a third inner sidewall SD**2** of the top electrode structure **140** to a fourth inner sidewall SD**2**′ of the top electrode structure **140**, and the upper portion Q**3** has a third width W**3** measured from a fifth inner sidewall SD**3** of the top electrode structure **140** to a sixth inner sidewall SD**3**′ of the top electrode structure **140**.

[0076] In some embodiments, the first width W**1** is greater than the second width W**2**. In some embodiments, the second width W**2** is greater than the third width W**3**. In some embodiments, the first width W**1** is greater than either the second width W**2** or the third width W**3**.

[0077] In some alternative embodiments, the third width W**3** is comparable to the second width W**2**, and the second width W**2** is comparable to the first width W**1**.

[0078] FIG. **25** is a cross-sectional diagram of an intermediate stage in the formation of a memory device in accordance with some embodiments of the present disclosure. In the operation S**22**, an air gap structure **211** is formed between the plurality of landing areas **102** by performing a thermal treating process. In some embodiments, during the thermal treating process, the remaining energy-removable layer **213** is transformed into the air gap structure **211** including an air gap **211**C enclosed by a liner **211**B.

[0079] In some embodiments, the air gap structure **211** is sealed by the first support layer **104**, and a portion of the first support layer **104** extends into the space between the plurality of landing areas **102**. In other words, the top surface of the air gap structure **213** is lower than the top surfaces **102**T of the plurality of landing areas **102** (not shown). The presence of the air gap structure **211** can reduce the parasitic capacitance between the plurality of landing areas **102**.

[0080] FIG. **26** is a flow diagram illustrating a method of manufacturing a memory device in accordance with some embodiments of the present disclosure. The method S**2** includes a number of operations (S**11**, S**12**, S**13**, S**14**, S**15**, S**16**, S**17**, S**18**, S**19**′, S**20**′, S**21**′, and S**22**′) and the description and illustration are not deemed as a limitation to the sequence of the operations. In the operation S**11**, a landing area is formed over a substrate, an energy-removable layer is formed adjacent to the landing area, and a first support layer is formed over the landing area. In the operation S**12**, a first material layer is formed over the first support layer, wherein the first material layer is doped with an N-type dopant. In the operation S**13**, a second material layer is formed over the first material layer. In the operation S**14**, a second support layer is formed over the second material layer. In the operation S**15**, a third material layer is formed over the second support layer, wherein the third material layer is doped with a P-type dopant. In the operation S**16**, a third support layer is formed over the third material layer. In the operation S**17**, a dry etch operation is performed to form a first recess, wherein the first recess penetrates the first material layer, the second material layer, and the third material layer. In the operation S**18**, a bottom electrode material layer is formed in the first recess. In the operation S**19′**, a portion of the bottom electrode material layer is removed to form a bottom electrode, and the first material layer, the second material layer and the third material layer are removed. In the operation S**20′**, a dielectric layer is conformally formed over the bottom electrode and a dielectric portion is formed on the dielectric layer and covering a top corner of the first recess. In the operation S**21′**, a top electrode is formed over the dielectric layer and the dielectric portion. In the operation S**22′**, an air gap structure is formed by performing a thermal treating process.

[0081] FIGS. **1** to **15** and **27** to **34** are schematic diagrams illustrating various fabrication stages constructed according to the method S**2** in accordance with some embodiments of the present disclosure.

[0082] After performing operations S**11** to S**18** as depicted in FIGS. **1** to **15**, operations S**19′** to S**22′** are performed, as will be depicted in FIGS. **27** to **34**.

[0083] FIG. **27** is a cross-sectional diagram of an intermediate stage in the formation of a memory device in accordance with some embodiments of the present disclosure, and FIG. **28** is a top view perspective schematic diagram of an intermediate stage in the formation of the memory device shown in FIG. **27** in accordance with some embodiments of the present disclosure. In the operation S**19′**, a cutoff region J**1** is defined by a lithography operation, wherein the cutoff region J**1** connects to at least two first recesses R**1**. In some embodiments, each cutoff region J**1** connects to three first recesses R**1**, as depicted in FIG. **28**. Further, an etching operation is performed to remove a portion of the third material layer **108**, a portion of the third support layer **109**, and an upper portion of the bottom electrode material layer **111**M (depicted in FIG. **15**) over the cutoff region J**1**. In some embodiments, the etching operation stops at a position in the third material layer **108**, or, alternatively stated, the etching operation stops at a position above a top surface **107**T of the second support layer **107**. After such etching operation, a remaining portion of the third material layer **108** in the cutoff region J**1** has a top surface **108**T**2** lower than the top surface **108**T**1** of the third material layer **108** that is outside of the cutoff region J**1**.

[0084] The bottom electrode material layer **111**M is separated into multiple portions after the etching operation, thereby forming multiple bottom electrodes **111** over each landing area **102**. For example, one bottom electrode **111** is formed over the first region RA, and another bottom electrode **111** is formed over the second region RB. Each bottom electrode **111** has a container-shaped profile.

[0085] The bottom electrode **111** includes a first wall portion **111**A extending upward and adjacent to the cutoff region J**1**; a second wall portion **111**B extending upward and positioned on a side away from the first wall portion **111**A; a bottom portion **111**C over the landing area **102**; and a top portion **111**D extending over the third support layer **109**.

[0086] The first wall portion **111**A has a top surface **111**T**2** lower than the top surface **108**T**1** of the third material layer **108**. The top surface **111**T**2** is lower than the top surface **111**T**1** of the top portion **111**D of the bottom electrode **111**. In some embodiments, the top surface **111**T**2** is level with the top surface **108**T**2**. A cleaning operation can be performed to remove residues generated during the etching operation.

[0087] FIG. **29** is a cross-sectional diagram of an intermediate stage in the formation of a memory device in accordance with some embodiments of the present disclosure. In the operation S**19′**, the third material layer **108** is removed by a removal operation. In some embodiments, the removal operation includes applying NF.sub.3 and H.sub.2, along with an application of plasma, over the third material layer **108**. The removal operation has an etch rate on Si significantly greater than an etch rate on SiN; for example, around 2000:1. Further, when the targeted layer to be etched is partially covered by SiN, the aforesaid etching recipe is an effective operation. After the removal operation is performed, the third support layer **109** remains adhered to the bottom electrode **111** and overhangs the second support layer **107**. In some embodiments, the third support layer **109** is attached to the second wall portion **111**B and the top portion **111**D of the bottom electrode **111**.

[0088] The top surface **107**T of the second support layer **107** is exposed, and an empty space E**1** is formed between the second support layer **107** and the third support layer **109**.

[0089] After the removal operation, a punch-through operation is performed to remove a portion of the second support layer **107** in the cutoff region J**1**. Therefore, a second recess R**2** is formed in the second support layer **107**, and a top surface **106**T of the second material layer **106** is exposed through the second recess R**2**. The second recess R**2** is positioned between the first wall portions **111**A of a plurality of the bottom electrodes **111**.

[0090] FIG. **30** is a cross-sectional diagram of an intermediate stage in the formation of a memory device in accordance with some embodiments of the present disclosure. In the operation S**19**′, the first material layer **105** and the second material layer **106** are removed by a removal operation. In some embodiments, the removal operation includes applying NF.sub.3 and H.sub.2, along with an application of plasma, over the first material layer **105** and the second material layer **106**. Such removal operation has an etch rate on Si significantly greater than an etch rate on SiN, for example, around 2000:1. After performing the removal operation, the second support layer **107** remains adhered to the bottom electrode **111** and overhangs the first support layer **104**. In some embodiments, the etching chemical enters through the second recess R**2** depicted in FIG. **29**. In some embodiments, the second support layer **107** is attached to the second wall portion **111**B of the bottom electrode **111**. The top surface **104**T of the first support layer **104** is exposed. An empty space E**2** is formed between the first support layer **104** and the second support layer **107**. An empty space E**3** is formed between the first wall portions **111**A of the plurality of bottom electrodes **111**.

[0091] In summary, in the operation S**19**′, a portion of the bottom electrode material layer **111**M is removed to form the bottom electrode **111**, and the first material layer **105**, the second material layer **106**, and the third material layer **108** are removed. The first support layer **104**, the second support layer **107**, the third support layer **109**, and the bottom electrodes **111** remain in place.

[0092] FIG. **31** is a cross-sectional diagram of an intermediate stage in the formation of a memory device in accordance with some embodiments of the present disclosure. In the operation S**20**′, a plurality of dielectric layers **112** are conformally formed over an exposed area of the bottom electrodes **111**. Each of the plurality of dielectric layers **112** includes a first part **112**A on a side of the bottom electrode **111** proximal to the first recess R**1**, a second part **112**B on an opposite side of the second wall portion **111**B (which is proximal to the empty space E**1** and the empty space E**2**), and a third part **112**C proximal to the empty space E**3**. Moreover, in some embodiments, the top corners TC of the plurality of first recesses R**1** are also covered by the plurality of dielectric layer **112**.

[0093] In some embodiments, each of the plurality of dielectric layers **112** includes a single layer or multiple layers. In some embodiments, each of the plurality of dielectric layers **112** includes SiO.sub.2, a dielectric material with high dielectric constant (high-k), such as ZrO.sub.2, HfO.sub.2, TiO.sub.2, AlO, or a combination thereof. For example, each of the plurality of dielectric layers **112** may be a tri-layer structure including two layers of aluminum oxide and a layer of zirconium oxide disposed between them. Moreover, in some embodiments, the plurality of dielectric layers **112** are formed by a conformal deposition process, such as an ALD process.

[0094] FIG. **32** is a cross-sectional diagram of an intermediate stage in the formation of a memory device in accordance with some embodiments of the present disclosure. In the operation S**20**′, a plurality of dielectric portions **157** are deposited over the plurality of dielectric layers **112** and at the top corners TC of the plurality of first recesses R**1**, respectively and correspondingly. In some embodiments, the plurality of dielectric layers **112** and the plurality of dielectric portions **157** are collectively referred to the plurality of dielectric structures **139**. It should be noted that the plurality of dielectric portions **157** are formed by a deposition process having a lower step coverage than the deposition process used for forming the plurality of dielectric layers **112**. In some embodiments, the plurality of dielectric portions **157** are formed by a non-conformal deposition process, such as a non-conformal liner atomic layer deposition (NOLA) process.

[0095] For brevity, clarity, and convenience of description, only one dielectric structure **139** is described. After the dielectric portion **157** is formed, a portion of the first part **112**A of the dielectric layer **112** are covered by the dielectric portion **157**. Detailedly, the first part **112**A of the dielectric layer **112** has a portion sandwiched between the dielectric portion **157** and the top portion **111**D and has another portion sandwiched between the dielectric portion **157** and the second wall portion **111**B. In some embodiments, the dielectric portion **157** includes silicon oxide, silicon nitride, silicon oxynitride, a dielectric material with high dielectric constant (high-k), or a combination thereof.

[0096] The dielectric portion **157** is formed to prevent the corner effect, which incur uneven thickness of the dielectric layer **112**. By forming the dielectric portion **157** over the dielectric layer **112**, the overall thickness of the dielectric structure **139** at the top corners TC of the first recess R**1** may be increased, which prevent or reduce leakage current between the plurality of bottom electrodes **111** and the subsequently formed top electrode.

[0097] FIG. **33** is a cross-sectional diagram of an intermediate stage in the formation of a memory device in accordance with some embodiments of the present disclosure. In the operation S**21**′, a plurality of top electrodes **113** are conformally formed over the plurality of dielectric structures **139**, respectively and correspondingly. In some embodiments, the plurality of top electrodes **113** may include TiN, and can be formed by an atomic layer deposition operation.

[0098] One set of the top electrode **113**, the bottom electrode **111**, and the dielectric structure **139** is collectively referred to as a container structure **147**. The container structure **147** can also be referred to as a double side container, wherein the top electrode **113** is formed over two sides of one bottom electrode **111**.

[0099] One container structure **147** has a lower portion P**1**′, a middle portion P**2**′ above the lower portion P**1**′, and an upper portion P**3**′ above the middle portion P**2**′. The first recess R**1** defined by one container structure **147** has a dimension D**1**′ at a level of the lower portion P**1**′, a dimension D**2**′ at a level of the middle portion P**2**′, and a dimension D**3**′ at a level of the upper portion P**3**′. In some embodiments, the dimension D**1**′ is greater than the dimension D**2**′. In some embodiments, the dimension D**2**′ is greater than the dimension D**3**′. In some embodiments, the dimension D**1**′ is greater than either the dimension D**2**′ or the dimension D**3**′. The dimensions D**1**′, D**2**′ and D**3**′ are maximum widths in a lateral direction of a corresponding empty space of the first recess R**1**.

[0100] In some alternative embodiments, the dimension D**1**′ is comparable to the dimension D**2**′, and the dimension D**2**′ is comparable to the dimension D**3**′.

[0101] Further, the lower portion P**1**′ has a first width W**1** measured from a first inner sidewall SW**1** of the bottom electrode **111** to a second inner sidewall SW**1**′ of the bottom electrode **111**, the middle portion P**2**′ has a second width W**2** measured from a third inner sidewall SW**2** of the bottom electrode **111** to a fourth inner sidewall SW**2**′ of the bottom electrode **111**, and the upper portion P**3**′ has a third width W**3** measured from a fifth inner sidewall SW**3** of the bottom electrode **111** to a sixth inner sidewall SW**3**′ of the bottom electrode **111**.

[0102] In some embodiments, the first width W**1** is greater than the second width W**2**. In some embodiments, the second width W**2** is greater than the third width W**3**. In some embodiments, the first width W**1** is greater than either the second width W**2** or the third width W**3**.

[0103] In some alternative embodiments, the third width W**3** is comparable to the second width W**2**, and the second width W**2** is comparable to the first width W**1**.

[0104] In some embodiments, a total thickness h**5** of the container structure **147** may be in a range

from 0.8 μm to about 1.2 μm. The first recess R**1** is laterally surrounded by the top electrode **113** and the bottom electrode **111**.
[0105] Optionally, another insulation layer can be formed over the top electrode **113**, but the present disclosure is not limited thereto.
[0106] As previously discussed, the container structure **147** has a profile that has a relatively wider bottom width (compared to a profile of the comparative embodiment depicted in FIG. **1**); thus, it has a greater capacitance, which improves a device performance of the memory device.
[0107] FIG. **34** is a cross-sectional diagram of an intermediate stage in the formation of a memory device in accordance with some embodiments of the present disclosure. In the operation S**22**′, an air gap structure is formed between the plurality of landing areas **102** by performing a thermal treating process. In some embodiments, during the thermal treating process, the remaining energy-removable layer **213** is transformed into an air gap structure **211** including an air gap **211**C enclosed by a liner **211**B.
[0108] In some embodiments, the air gap structure **211** is sealed by the first support layer **104**, and a portion of the first support layer **104** extends into the space between the plurality of landing areas **102**. In other words, the top surface of the air gap structure **213** is lower than the top surfaces **102**T of the plurality of landing areas **102** (not shown).
[0109] One aspect of the present disclosure provides a memory device including a substrate; a landing area positioned on the substrate; a bottom electrode positioned on the landing area, wherein the bottom electrode has a container-shaped profile; a support layer positioned over the substrate and laterally surrounded the bottom electrode; a dielectric structure including a dielectric layer conformally positioned on the bottom electrode and on a top surface of the support layer, and covering top corners of the support layer, and a plurality of dielectric portions conformally positioned on the dielectric layer and covering the top corners of the support layer; and a top electrode structure positioned on the dielectric structure. The plurality of dielectric portions are sandwiched by the top electrode structure and the dielectric layer. The top surface of the third support layer is higher than a top surface of the bottom electrode.
[0110] Another aspect of the present disclosure provides a memory device including a substrate; a landing area positioned on the substrate; a first support layer positioned on the substrate; a second support layer positioned over the first support layer; a third support layer positioned over the second support layer; a bottom electrode including a bottom portion horizontally positioned on the landing area, a first wall portion extending from the bottom portion along a first direction, a second wall portion extending from the bottom portion along the first direction and separated from the first wall portion, and a top portion connecting to the second wall portion, parallel to the bottom portion, and collectively covering a top corner of the third support layer with the second wall portion; a dielectric structure including a dielectric layer conformally positioned on the bottom electrode, and a dielectric portion positioned on the dielectric layer and covering the top corner of the third support layer; and a top electrode conformally positioned on the dielectric structure.
[0111] Another aspect of the present disclosure provides a method for fabricating a memory device including forming a landing area on a substrate; forming an energy-removable layer adjacent to the landing area; sequentially forming a first support layer, a first material layer, a second material layer, a second support layer, a third material layer, and a third support layer over the landing area, wherein the first material layer is doped with an N-type dopant and the third material layer is doped with a P-type dopant; forming a first recess to expose the landing area; forming a bottom electrode within the first recess; conformally forming a dielectric layer on the bottom electrode and on a top surface of the third support layer; forming a plurality of dielectric portions on top corners of the first recess; and conformally forming a top conductive layer on the dielectric layer and covering the plurality of dielectric portions; and forming a top electrode on the top conductive layer.
[0112] Due to the design of the memory device of the present disclosure, the overall thickness of the dielectric structure **139** at the top corner TC of the third support layer **109** is increased. As a

result, the leakage current between the bottom electrode **111** and the top electrode structure **140** can be prevented or reduced. In addition, the first width W**1** of the lower portion Q**1** of the top electrode structure **140** is greater than the third width W**3** of the upper portion Q**3** of the top electrode structure **140**. Consequently, the capacitance of the capacitor structure **148** can be improved. Furthermore, the air gap structure **211** positioned between the plurality of landing areas **102** can reduce the parasitic capacitance between the plurality of landing areas **102**.
[0113] Although the present disclosure and its advantages have been described in detail, it should be understood that various changes, substitutions and alterations can be made herein without departing from the spirit and scope of the disclosure as defined by the appended claims. For example, many of the processes discussed above can be implemented in different methodologies and replaced by other processes, or a combination thereof.
[0114] Moreover, the scope of the present application is not intended to be limited to the particular embodiments of the process, machine, manufacture, composition of matter, means, methods and steps described in the specification. As one of ordinary skill in the art will readily appreciate from the disclosure of the present disclosure, processes, machines, manufacture, compositions of matter, means, methods, or steps, presently existing or later to be developed, that perform substantially the same function or achieve substantially the same result as the corresponding embodiments described herein may be utilized according to the present disclosure. Accordingly, the appended claims are intended to include within their scope such processes, machines, manufacture, compositions of matter, means, methods, and steps.

Claims

1. A method for fabricating a memory device, comprising: forming a landing area on a substrate; forming an energy-removable layer adjacent to the landing area; sequentially forming a first support layer, a first material layer, a second material layer, a second support layer, a third material layer, and a third support layer over the landing area, wherein the first material layer is doped with an N-type dopant and the third material layer is doped with a P-type dopant; forming a first recess to expose the landing area; forming a bottom electrode within the first recess; conformally forming a dielectric layer on the bottom electrode and on a top surface of the third support layer; forming a plurality of dielectric portions on top corners of the first recess; and conformally forming a top conductive layer on the dielectric layer and covering the plurality of dielectric portions; and forming a top electrode on the top conductive layer.
2. The method for fabricating the memory device of claim 1, wherein the first support layer, the second support layer, and the third support layer comprise silicon nitride.
3. The method for fabricating the memory device of claim 2, wherein the first material layer, the second material layer, and the third material layer comprise polysilicon.
4. The method for fabricating the memory device of claim 2, a dimension of the first recess at a level of the first material layer is greater than a dimension of the first recess at a level of the third material layer.
5. The method for fabricating the memory device of claim 1, further comprising performing a thermal treating process to turn the energy-removable layer into an air gap structure comprising an air gap enclosed by a liner.
6. The method for fabricating the memory device of claim 5, wherein the energy-removable layer comprises a thermal decomposable material, a photonic decomposable material, or an e-beam decomposable material.